Active Challenge-Response Biometric Verification for Video KYC via Speech Aerodynamics and Trellis Forced Alignment

Jatin Dhawad
Department of Computer Engineering
School of Technology Management & Engineering (STME)
SVKM's NMIMS, Indore, India
jatindhawad22@gmail.com

Param Rawal
Department of Computer Engineering
School of Technology Management & Engineering (STME)
SVKM's NMIMS, Indore, India
paramrawal290@gmail.com

***Abstract*—**Onboarding with remote Video KYC is vulnerable to synthetic identity injection, because virtual cameras and real-time face re-enactment can produce clean frames that may escape passive attack detection. In this paper we introduce SecureKYC AI, an active challenge-response approach based on one physical property of human speech: articulating bilabial consonants (/m/, /b/, /p/) requires the lips to be closed ($P_{oral} >> P_{ambient}$). The server delivers a randomly generated challenge phrase. MediaPipe Face Mesh tracks lip movement during articulation of the phrase, wav2vec2 forced alignment (MMS_FA) marks the moment of each bilabial occurrence, and a scale-independent 3D lip aperture is measured at these timestamps. On N=691 audio-visual clips (43 disjoint subjects; train/validation/test = 25/9/9 identities and 484/104/103 clips), the deployed constrained late-fusion meta-learner achieves 68.93% accuracy, 0.7197 test ROC-AUC (5-fold CV: 0.6877 ± 0.0449), 36.02% EER, and 0.6000 F1-score. At the default 0.5 operating point, late fusion produced 10 false alarms on the 57 genuine test clips versus 16 for the standalone vision branch (paired bootstrap 95% CI for the difference [0, 12]; the interval reaches zero), with identical recall (52.17%) and higher precision (70.59% vs 60.00%); at the equal-error point the two models are identical. Across all 691 clips, the closure-failure ratio averages 4.5% for authentic and 41.6% for reenacted clips (Cohen's d of S_{desync}: 0.66 per clip, 1.10 per subject). The average processing time for each frame offline at steady-state is 5.02 ms (14.05 ms at the start-up phase, according to nine video demonstrations). This research focuses only on in-distribution benchmark tests, which require further validation in practice using multi-camera test scenarios.

***Index Terms*—**Video KYC, Deepfake Detection, Presentation Attack Detection, Speech Aerodynamics, Forced Alignment, Bilabial Closure, Active Challenge-Response, Multimodal Biometrics.

I. INTRODUCTION

Onboarding of remote users in accordance with RBI Master Directions on KYC and Europe's AMLD5 and eIDAS rules requires a WebRTC video call session. Recent work on facial biometric security has proposed multimodal deep learning with cryptographic protection to guard facial authentication against spoofing [22]. The paradigm assumes a live applicant positioned in front of an optical camera sensor. However, this assumption is negated by modern real-time generative models. Traditional presentation attacks, which include printed images, silicone masks, or screen playback, required traversal of the optical path. In the case of today's attacks, synthetic frames can be injected directly into the input stream [1], [2].

Optical liveness detection techniques consider imaging artifacts such as frequency-domain artifacts [3], up-convolution effects [4], pose drifts [5], and rPPG heartbeats [6], [7]. Injection attacks remove the optical path from the threat model. Loopback drivers like OBS Virtual Camera, ManyCam, and v4l2loopback for Linux enable arbitrary image buffers to be used as input device streams. A browser sandbox has no way to check driver signatures, so the distant verifier gets a clean pixel image without any glare, moire effect, or distortion effect that usually happens in physical capturing process.

The real-time puppetry makes the injection interactive. The attacker sits at a webcam and speaks, and a reenactment tool such as DeepFaceLive (built on DeepFaceLab) tracks their facial keypoints frame by frame and renders the victim's face over those movements [8]. Passive presentation attack detection (PAD) in the sense of ISO/IEC 30107-3 does not stop this. These blink, smile, and yaw triggers fail since the attacker carries out the trigger, and the neural mask simulates it with sub-frame latency. Behavioral biometrics, including continuous person authentication, have been surveyed for identity verification in domains such as banking and finance [23]. This entire process is shown in Fig. 1.

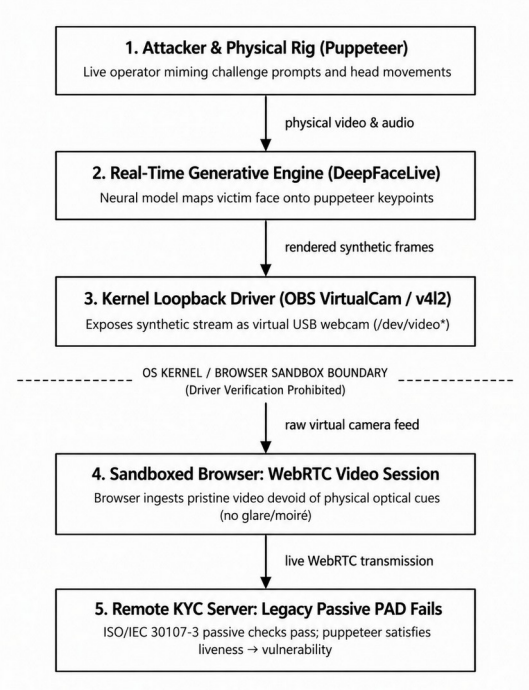


Fig. 1. Video KYC – virtual camera injection threat model. The attack consists of injecting a fake video during the video call to evade the optical liveness check.

A. Prior Art: The Limitations of Visual Challenge Cascades

Passive mechanisms are not effective for this, and therefore, a move toward the active challenge-response mechanisms is called for. The most relevant previous mechanism is GOTCHA by Mittal et al. (2024), presented at IEEE EuroS&P [9]. GOTCHA uses physical challenges; it involves either covering the face using the hand, standing up, or moving the head so as to change the yaw angle. Generally, the principle behind this is that generators will fail to accomplish this because of hand occlusion and the change in yaw angle. The mechanism obtains an automatic AUC of 80.1% (88.6% human AUC) [9].

Adapting this mechanism to a Video KYC setting leads to three major issues:

1) User effort. It will be challenging to ask the user to perform actions such as covering their face with a hand or standing up during a formal onboarding identity call.
2) Time consumption. Physical challenges take time, which contradicts the requirement of a sub-second reaction time after the utterance.
3) No voice is used. The GOTCHA system works solely with the video feed and does not consider either the voice input or the actual physical process that produces the voice.

B. Contributions of This Work

SecureKYC AI is a liveness-based approach, which leverages speech generation physical features. There are five main points that could be highlighted:

1) Live test using aerodynamics. By definition, liveness relies on aerodynamics of speech (P_{oral} >> $P_{ambient}$). The presence of bilabial phonemes /m/, /b/, and /p/ demands closure of the lips, which is usually absent in face-swapped videos from the dataset.
2) Dynamic Trellis Forced Alignment without Blind Automatic Speech Recognition (ASR). The challenge phrase is selected by the server from a list of predefined phrases containing bilabial phonemes (e.g., "My Bank Passcode"), thereby allowing the ground truth transcription of the utterance to be obtained even prior to saying it. MMS_FA trellis alignment then extracts the timing of every bilabial articulation without any open-vocabulary speech recognition.
3) Dual-Budget Real-Time Architecture. We split the computation into a streaming visual tracking pass (the offline endpoint's amortized per-frame time, covering video decoding, landmark tracking, vision-branch inference and the audio check, was 5.02 ms in steady state over 9 demo clips and 14.05 ms on the first call) and a post-utterance acoustic alignment pass, giving a client-side verification time of 224.3 ± 21.7 ms on Apple Silicon MPS (local in-process client).
4) Constrained Late-Fusion Meta-Learner. We found feature scale disparity and gradient interference in early-fusion networks (Section V-B), so we train the two branches separately and combine them with a non-negatively constrained stacking meta-learner ($W_{vis} \geq 0$, $W_{desync} \geq 0$). The constraint prevented the weight sign inversion we saw in one cross-validation fold.
5) Strict Empirical Verification. Adhering strictly to zero-synthetic-data and zero-identity-leakage protocols, we evaluate the system on N=691 clips across 43 disjoint subjects (25 train, 9 val, 9 test identities; 484/104/103 clips), validating an operating accuracy of 68.93% and 0.7197 test ROC-AUC.

II. AERODYNAMIC & BIOMECHANICAL FOUNDATIONS

A. Fundamentals of Bilabial Speech Aerodynamics

Human speech production is governed by the continuous mechanical manipulation of pulmonary airflow across the vocal tract [10], [11]. Consonants are classified by their manner and place of articulation. Among English phonemes, the bilabial consonants—the voiced bilabial nasal /m/, voiced bilabial stop /b/, and voiceless bilabial stop /p/—possess a unique biomechanical invariant: complete, airtight mechanical coaptation of the vermilion borders of the upper lip (labium superius) and lower lip (labium inferius) [11]. Fig. 2 compares genuine and reenacted bilabial production.

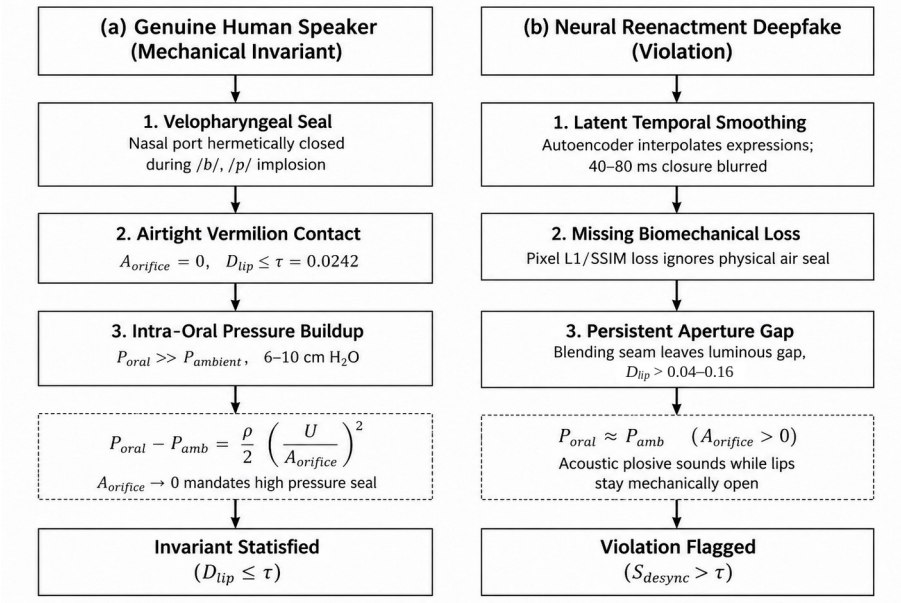


Fig. 2. Aerodynamic and biomechanical comparison of bilabial consonant production (/m/, /b/, /p/) between genuine human speakers and neural reenactment deepfakes.

During the closure phase (implosion) of /b/ and /p/, the velopharyngeal port closes and seals off the nasal cavity. Pulmonary airflow continues into the closed oral cavity, so intra-oral pressure (P_{oral}) rises quickly above ambient atmospheric pressure ($P_{ambient}$). According to Bernoulli's acoustic volume velocity relation [10]:

$$P_{oral} - P_{ambient} = (\rho / 2) \cdot (U_air / A_orifice)^2 \quad (1)$$

In this relation ρ is air density, U_air is the acoustic volume velocity, and A_orifice is the cross-sectional area of the lip opening. Equation (1) holds only while the orifice is open. A burst release needs the lips to seal first (A_orifice → 0): the seal stops the airflow through them, so P_{oral} builds up behind the closure to a peak of about 6 to 10 cm H2O above atmospheric pressure [11]. For /m/, the velopharyngeal port is open, but the lips stay sealed, so the sound leaves only through the nose.

B. Why Generative Face-Swaps Fail at Bilabial Closure

Neural face reenactment and face-swap models (autoencoders, GANs, latent diffusion) have three built-in limits that make them likely to get bilabial closure wrong:

1) Latent averaging. These models interpolate expressions in a low-dimensional latent space. Going from an open vowel to a closed bilabial takes only 40 to 80 ms (1 to 3 frames at 30 FPS). Temporal smoothing and the encoder-decoder bottleneck blur such fast movement, so the synthetic lips can stay partly open [12], [13].

2) No physical constraint. Common deepfake losses (perceptual L1/L2, multi-scale SSIM, PatchGAN adversarial loss) reward pixel similarity. None of them penalizes a missing air seal [12].

3) Blending artifacts. The mouth interior is full of fine detail (teeth, tongue, shadow), and generators often leave blending seams along the lip edge, which can leave a bright gap between the upper and lower lip [14].

Fig. 3 demonstrates the difference. Real speakers consistently produce normalized lip aperture values lower than τ = 0.0242 for bilabials. For the swapped samples shown in the figure, D_{lip} at the bilabial centre (t = 0) varies between 0.04 and 0.16, which is higher than τ in all six traces, although one trace reaches about τ at +40 and +80 ms.

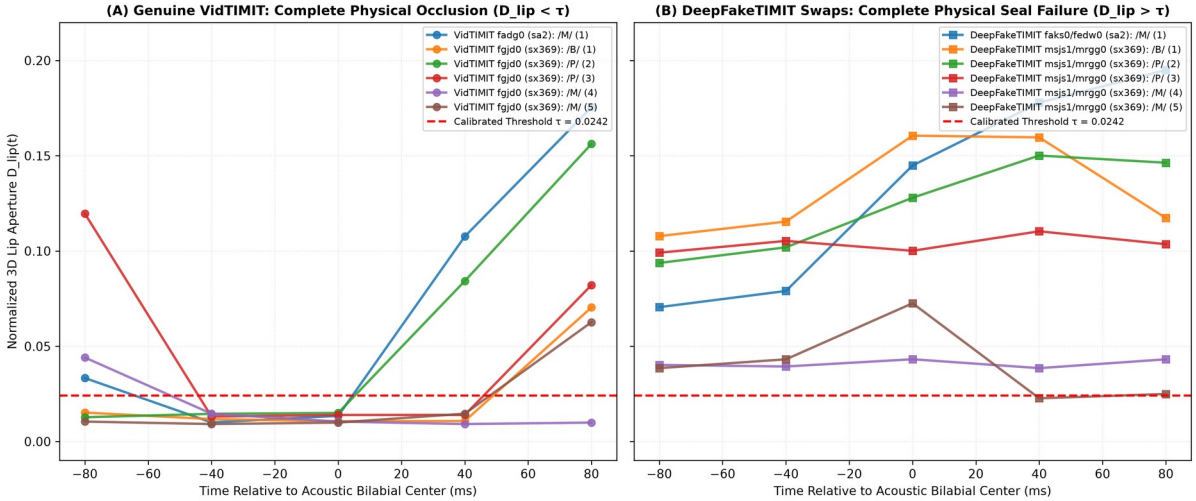


Fig. 3. Normalized 3D lip aperture $D_{lip}(t)$ for real VidTIMIT speakers (Panel A: fadg0, fgjd0) and DeepFakeTIMIT swaps (Panel B: faks0/fedw0, msjs1/mrgg0), plotted with scripts/plot_bilabial_window_trajectories.py. Real speakers drop below τ = 0.0242 at the forced-alignment bilabial times $T_{bilabial}$. In the swapped clips the mouth mostly stays open.

III. SECUREKYC AI SYSTEM ARCHITECTURE

The overall architecture of the system is shown in Fig. 4. SecureKYC AI operates on two separate streams: a visual stream that runs in real time as the speaker talks, and an acoustic alignment procedure that runs after the utterance ends.

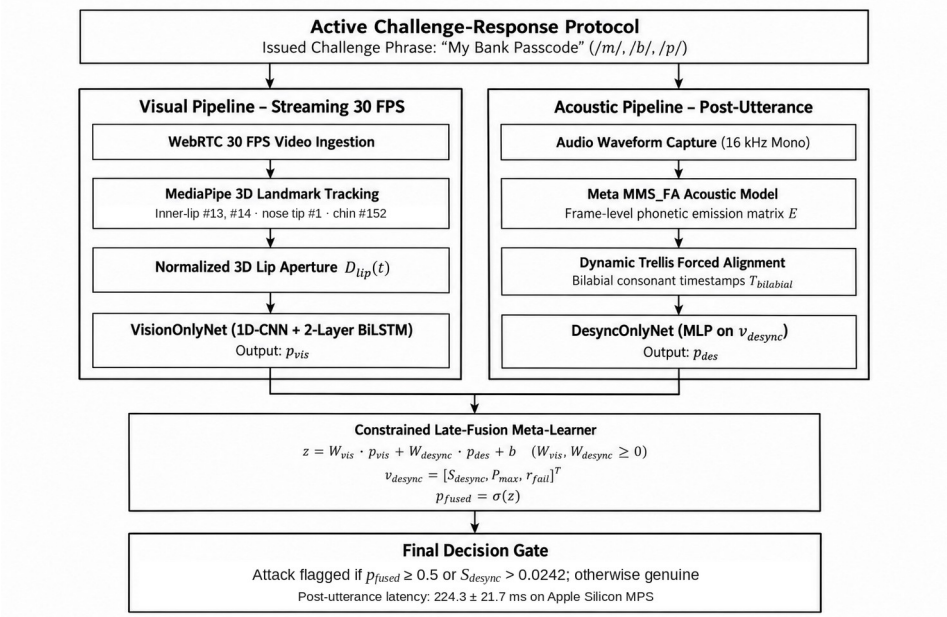


Fig. 4. Overview of the SecureKYC AI system. Visual tracking runs during streaming, and audio alignment and scoring run after the utterance.

A. Real-Time 3D Lip Aperture Metric

The lip aperture is computed independently of camera distance by tracking 3D facial landmarks with the MediaPipe Face Mesh [15] library, with landmark refinement enabled and the face-detection confidence threshold set to 0.5. Facial analysis in the wild, where head movement, face deformation, and illumination changes degrade features, has been studied with residual networks [24]. Let p_i = (x_i, y_i, z_i) be the coordinates of the i-th landmark. If face detection fails for several consecutive frames, the missing apertures are linearly interpolated, provided the gap is no longer than three frames. The interpolated apertures are used by VisionOnlyNet, while the desynchronization score takes into account only the frames where the face was detected. Clips with longer gaps are considered to have an unreliable facial track.

The normalized lip aperture $D_{lip}(t)$ in video frame t is defined as the distance between the inner upper and lower lip points divided by the face height (the distance between the nose tip and the chin):

$$D_{lip}(t) = ||p_{13}(t) - p_{14}(t)||_2 \;/\; (||p_1(t) - p_{152}(t)||_2 + 10^{-6}) \qquad (2)$$

Here p_{13} is the center of the inner upper lip, p_{14} is the center of the inner lower lip, p_1 is the nose tip, and p_{152} is the chin landmark, which means that the denominator is equal to the face height. The

constant 10^{-6} prevents division by zero. Normalizing by face height makes $D_{lip}(t)$ almost independent of camera distance and downsampling.

B. Trellis Forced Alignment via MMS_FA

Most audio-visual sync methods transcribe the speech first, and recognition errors on short, noisy recordings then propagate into the alignment. In SecureKYC AI the server generates the challenge prompt, so the intended transcript is known in advance. Before any alignment is attempted, a separate character-error-rate (CER) check (Section V-E) using a wav2vec2 CTC recognizer rejects audio that does not match the issued prompt.

Let a = [a_1, a_2, ..., a_T] denote the resampled 16 kHz audio waveform. We pass a through the pre-trained Meta MMS_FA wav2vec2 acoustic model [16], [17] to obtain the frame-level emission log-probabilities E, a matrix of size $T_{audio} \times |V|$, where |V| is the model's character vocabulary (26 lowercase letters, an apostrophe, a word-boundary token and the CTC blank; 29 labels in total).

We build an alignment trellis α(t, j) giving the log-probability of aligning acoustic frame t to character token j of the known phrase. Because the tokens are letters, we treat the tokens m, b and p as bilabial events, which is reliable only because the challenge phrases come from a controlled pool:

$$\alpha(t, j) = max(\alpha(t-1, j), \alpha(t-1, j-1)) + E(t, c_j) \quad (3)$$

The start and end times of each bilabial sound k in the set {/m/, /b/, /p/} can be expressed as:

$$T_{bilabial}^{(k)} = [t_{start}^{(k)}, \ t_{end}^{(k)}] \quad (4)$$

C. Cross-Modal Desynchronization Formulation

The frame closest to the center of the time-aligned interval of each bilabial closure k, among the frames where a face was detected, is selected and denoted $f_{mid}^{(k)}$. As anticipatory coarticulation enables lip closure to happen within a few tens of milliseconds before or after the burst release [11], a two-frame window around $f_{mid}^{(k)}$ is employed (±80 ms at the 25 fps of VidTIMIT):

$$W_k = [f_{mid}^{(k)} - 2, \ f_{mid}^{(k)} + 2] \quad (5)$$

Within this window, we check whether the speaker's normalized aperture falls below the physical closure threshold τ = 0.0242:

$$P_{sync}^{(k)} = max(0, \ min_{t \in Wk} D_{lip}(t) - \tau) \quad (6)$$

When there is no detected face in any frame within the window, the minimum aperture of the event is set to 1.0, meaning that the closure is considered a failure. From all K bilabial events in the utterance, we construct a three-dimensional cross-modal desynchronization vector $v_desync \in R^3$:

$$v_desync = [S_{desync}, \ P_{max}, \ r_{fail}] \quad (7)$$

where $S_{desync} = (1/K) \Sigma P_{sync}^{(k)}$, $P_{max} = \max_k P_{sync}^{(k)}$, and $r_{fail} = (1/K) \Sigma I(P_{sync}^{(k)} > 0)$.

D. Constrained Late-Fusion Meta-Learner

In our early experiments with a joint multi-task network (MultimodalKYCFusionNet), concatenating v_desync with the visual embeddings produced destructive gradient interference. We therefore split the architecture into two independently trained base classifiers:

1) Visual Base Network (VisionOnlyNet): A 1D convolutional feature extractor coupled to a 2-layer Bidirectional LSTM (hidden dimension 64) that processes the temporal aperture trajectory $D_{lip} \in R^{T \times 1}$, outputting posterior probability P_{vis} in [0, 1].
2) Acoustic-Phonemic Base Network (DesyncOnlyNet): A multi-layer perceptron processing v_desync, outputting posterior probability P_{desync} in [0, 1].

We combine the base predictions via a constrained late-fusion meta-learner:

$$z = W_{vis} \cdot P_{vis} + W_{desync} \cdot P_{desync} + b, \quad P_{fused} = \sigma(z) \quad (8)$$

Crucially, we enforce non-negativity constraints during optimization:

$$minimize \ LBCE(y, \sigma(z)) + (\lambda/2)(Wvis^2 + Wdesync^2) \ over \ Wvis, Wdesync, b$$
$$subject \ to \ \ Wvis \geq 0, \ \ Wdesync \geq 0 \quad (9)$$

solved with L-BFGS-B. Constraining both weights to be non-negative stops either branch from flipping sign on outlier speakers. It does not guarantee that fusion matches the better single branch (Table II, Fold 1).

IV. EXPERIMENTAL SETUP & PROTOCOL

A. Zero-Synthetic-Data & Hardware Target

This study uses no synthetic data and no simulated distributions. Every metric, ROC curve and latency figure we computed comes from the real signal-processing pipeline and live PyTorch forward passes; the exceptions are the GOTCHA AUC quoted from the literature and the dagger-marked Sep 4 ablation rows in Table I. Training and latency profiling were done using an Apple Silicon Mac, with Metal Performance Shaders (torch.device("mps")).

B. Identity-Disjoint Benchmark Dataset

With a random split at clip level, an individual can appear in both training and test splits, resulting in the problem of identity leakage [2], [13]. Identity leakage makes the classifier favor facial appearance over manipulation cues. The experiments employ N=691 video-audio clips: N=275 real videos from VidTIMIT [1] and N=416 fake videos generated by neural autoencoders in DeepFakeTIMIT [1].

Disjoint identity splits are made using split_dataset_by_identity(). This function constructs a graph over the dataset and does a connected component analysis to split the 43 identities into disjoint subsets:

- Training set: 25 identities (N=484 clips: 172 real and 312 fake).
- Validation Set: 9 identities (N=104 clips: 46 real, 58 fake).
- Held-Out Test Set: 9 completely unseen identities (N=103 clips: 57 real, 46 fake; Train ∩ Test = ∅).

The 9 held-out test identities are: fadg0, fcmr0, fpkt0, fram1, mbdg0, mbjk0, mreb0, mrjo0, and mstk0.

C. Physical Calibration of Closure Threshold

The closure detection threshold τ is derived from real VidTIMIT data as follows: N=320 bilabial closures by 15 speakers, calculated using src/metrics/calibrate_threshold.py and stored in data/processed/calibration_events_breakdown.json. The empirical distribution of D_{lip} at these events was:

- 95th Percentile (Calibrated Threshold): 0.0241 (~0.0242 in config.py)

According to definition, 95.0% of the 320 calibration measurements are ≤ τ = 0.0242. Across all 691 clips, the average per-clip closure-failure ratio is 4.49% for authentic clips and 41.63% for reenacted clips. The effect size of S_{desync} between reenacted and authentic clips is Cohen's d = 0.66 per clip and 1.10 per subject (42 authentic and 32 reenacted subjects). Note that the 15 calibration speakers include two of the nine test identities (fadg0 and fcmr0; 42 of the 320 events), so τ is not fully independent of the test split.

TABLE I: COMPARISON OF PERFORMANCE FOR MODEL ABLATIONS (TEST SET WITH HELD-OUT IDENTITY, N=103)

Model Version / Feature Options	Accuracy (%)	APCER (%)	BPCER (%)	Precision (%)	Recall (%)	F1-Score	ROC-AUC	Equal Error Rate (EER)
VisionOnlyNet (Spatial Standalone)	63.11%	47.83%	28.07%	60.00%	52.17%	0.5581	0.7105	36.02%
DesyncOnlyNet (Passive Standalone)*	44.66%	0.00%	100.00%	44.66%	100.00%	0.6174	0.6175	38.08%
Proposed Production: Constrained Late Fusion (Official Deployed)	68.93%	47.83%	17.54%	70.59%	52.17%	0.6000	0.7197	36.02%
Deployed Pipeline Rule: Constrained Late Fusion OR $S_{desync} > 0.0242$	68.93%	47.83%	17.54%	70.59%	52.17%	0.6000	—	—
Spatial-Only (Zero out desync_vec)†	**61.17%**	**—**	**—**	**68.75%**	**23.91%**	**0.3548**	**0.6182**	**—**
Passive-Only (Zero out lip_seq)†	55.34%	—	—	0.00%	0.00%	0.0000	0.6232	—
Baseline Early Fusion: MultimodalKYCFusionNet (Raw Concat)†	65.05%	—	—	67.86%	41.30%	0.5135	0.6423	—
Expected Generalization: Constrained Late Fusion (5-Fold CV Mean)	—	—	—	—	—	—	0.6877 ± 0.0449	36.00 ± 2.98%
*Literature Benchmark: GOTCHA (Mittal et al., 2024) [9]**	—	—	—	—	—	—	*0.8010**	—

**Model Rows vs. Deployed Pipeline Rule: The model rows are scored against a threshold of 0.5, except for ROC-AUC and EER, where all thresholds are considered. The Deployed Pipeline Rule adds the desynchronization override implemented in app/main.py (fused >= 0.5 OR $S_{desync} > 0.0242$). On the held-out test split (N=103), exactly 0 test clips satisfied $S_{desync} > 0.0242$ and $p_{fused} < 0.5$ (all 4 clips with $S_{desync} > 0.0242$ already had $p_{fused} \geq 0.5$); thus, the physical desynchronization override changed no test verdict on this split. At threshold 0.5, APCER (47.83%, 22/46) is identical for both VisionOnlyNet and Constrained Late Fusion. BPCER is 28.07% (16/57) for VisionOnlyNet versus 17.54% (10/57) for Constrained Late Fusion (paired bootstrap 95% CI for the difference [0, 12] false alarms; the interval reaches zero). Standalone DesyncOnlyNet collapses to predicting every clip as an attack (BPCER 100.00%, 57/57; APCER 0.00%, 0/46).*
**Note on GOTCHA: Literature-reported automated AUC in [9] across multi-stage visual cascades (hand occlusions, head turns). Evaluated on live streams in [9] and not trained or tested on this project's dataset; included strictly as a literature reference for active challenge-response protocols.*
†Logged ablation values from the Sep 4 training run. The standalone Bi-LSTM logged in that run reached AUC 0.7040 (Acc 64.08%, Prec 76.47%, Rec 28.26%, F1 0.4127); it came from a different training run than the production VisionOnlyNet baseline (AUC 0.7105). Neither EER nor operating thresholds are reported for the zero-masked feature ablations.

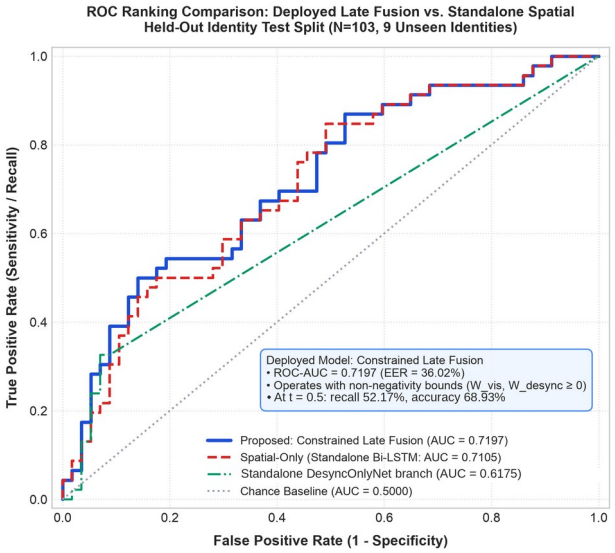


Fig. 5. *Receiver Operating Characteristic (ROC) curves on held-out test split (N=103). Deployed VisionOnlyNet achieves AUC = 0.7105, Deployed Constrained Late Fusion achieves AUC = 0.7197, and the green curve represents the standalone DesyncOnlyNet branch (AUC = 0.6175).*

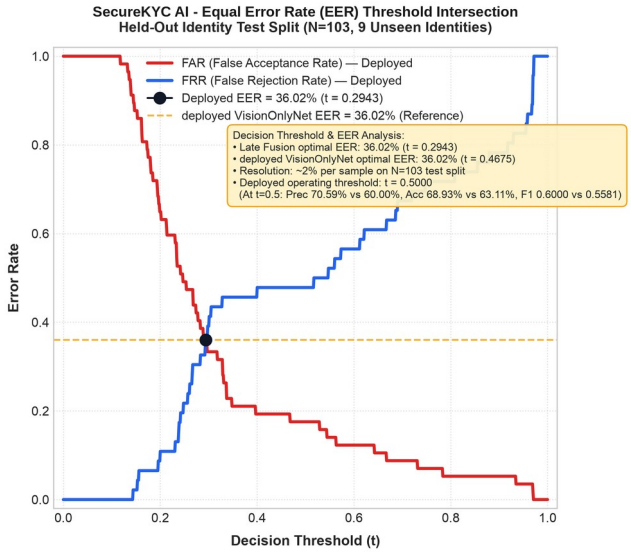


Fig. 6. *FAR and FRR curves. Deployed VisionOnlyNet and Constrained Late Fusion both have EER = 36.02% (resolution ~2% on 57/46 test clips; Late Fusion t = 0.2943, VisionOnlyNet t = 0.4675).*

V. EMPIRICAL RESULTS & DISCUSSION

A. Model Ablation Analysis

As Table I shows, at the default threshold (t=0.50) VisionOnlyNet and Constrained Late Fusion have identical recall (52.17%, 24 of 46 deepfakes detected). Late fusion cuts false alarms from 16 to 10 on the 57 genuine clips, which raises precision from 60.00% to 70.59%, accuracy from 63.11% to 68.93%, and F1-score from 0.5581 to 0.6000. The paired bootstrap 95% CI for the false-alarm difference is [0, 12], so this gain is suggestive rather than conclusive, and at the equal-error point the two models are identical. Both models achieve an identical EER of 36.02% (with thresholds of t = 0.4675 and t = 0.2943, both yielding a false acceptance rate (FAR) of 35.09% and a false rejection rate (FRR) of 36.96%). Fig. 5 shows the ROC curves and Fig. 6 shows the FAR and FRR curves behind these numbers. Ranking discrimination (AUC 0.7197 vs. 0.7105; paired bootstrap difference +0.0092, 95% CI [-0.0265, +0.0425] spanning zero) is comparable and within run-to-run sampling variation. The 5-fold CV mean AUC for late fusion (0.6877 ± 0.0449) is lower than standalone (0.7104). With 57 genuine and 46 deepfake test clips, each sample misclassification shifts error rates by approximately 2%, establishing the discrete resolution of the test partition.

B. Feature Scale Disparity and Late Fusion

The most important negative finding of this work is that early multimodal concatenation failed. In MultimodalKYCFusionNet, concatenating v_desync with the visual embeddings dropped test ROC-AUC from 0.7105 (Spatial Standalone) to 0.6423. The decoupled sensitivity analysis shows that the underlying cause is as follows:

- $S_{desync} \approx 10^{-2}$, whereas $r_{fail} \approx 10^{0}$.
- The linear projection weights have comparable values ($|W| \approx 0.05$), and therefore the pre-activation of r_{fail} becomes 60–100 times greater than that of S_{desync}, which effectively suppresses the contribution of the continuous metric penalty.
- When splitting modality to two base classifiers and then using non-negative constraint for late fusion (W_{vis} = 5.3688, W_{desync} = 4.9111, b = -6.0411), gradient interference is avoided.

TABLE II: 5-FOLD IDENTITY-GROUPED STABILITY OF THE META-LEARNER (34 TRAIN+VAL IDENTITIES, 588 CLIPS; AUC AND EER ON THE FIXED TEST SET, N=103)

Fold	Val IDs	Val N	W_{vis}	W_{desync}	Intercept	Spatial Standalone AUC	Constrained Late AUC	Δ vs. Spatial	EER (%)	Unconstrained Late AUC
1	7	119	0.0000	12.5037	-6.0913	0.7941	0.6175	-0.1766	38.08%	0.4613 (Crash)
2	7	113	4.5942	9.7904	-8.1186	0.7540	0.7548	+0.0008	30.13%	0.7548
3	7	125	2.6426	6.6113	-5.0020	0.6430	0.6758	+0.0328	37.99%	0.6758
4	7	108	6.2062	4.3486	-5.7960	0.6613	0.6812	+0.0198	36.90%	0.6812
5	6	123	4.2882	7.8638	-6.9213	0.6995	0.7094	+0.0099	36.90%	0.7094
Mean	—	—	**3.5463**	**8.2236**	**-6.3858**	**0.7104**	**0.6877**	**-0.0227**	**36.00%**	**0.6565**
Std	—	—	**2.1032**	**2.7755**	**1.0620**	**0.0565**	**0.0449**	**0.0777**	**2.98%**	**0.1121**

As can be seen from Table II, the non-negativity constraint is important. For instance, in the first fold, the unconstrained model gives a negative value of W_{vis} (-0.0111), while the test AUC is 0.4613, that is, below chance. With the non-negativity constraint, W_{vis} was clipped to 0.0000 and the model reverted to the desynchronization baseline (0.6175). Nevertheless, in Fold 1 the constrained model (0.6175) still performed worse than the standalone spatial branch (0.7941), which indicates that the constraint prevents inversion of the branches but does not ensure parity. It is critical to observe how the folds are constructed: for each fold, the base models are trained again, while the meta-learner is trained on the held-out fold (Val N). AUC and EER are then evaluated on the same fixed test split. The variability thus reflects differences in training sets rather than independent test sets.

C. Real-Time Operational Latency Budget

For automated Video KYC, we set our own target of a sub-second post-utterance response. We profile SecureKYC AI's computational latency across its dual-budget pipeline:

1) Streaming Visual Budget: The offline endpoint's amortized per-frame time (total time for video decoding, landmark tracking, vision-branch inference, audio extraction and the CTC prompt check, divided by the frame count) was 5.02 ± 0.12 ms in steady state over 9 demo clips and 45 timed runs (per-clip means 4.94–5.19 ms), and 14.05 ms on the first call. This is about 15% of the 33.3 ms frame window at 30 FPS (13% of the 40 ms window at the 25 fps of our demo clips); because it also includes the audio steps, it is an upper bound on the per-frame tracking cost. These timings come from the offline endpoint; we did not measure frame drops during a live stream.
2) Post-Utterance Verification Latency: Measured through the WebSocket verification path in app/main.py on 10 demo clips (6 genuine, 4 reenacted), with 2 discarded warm-up sessions and 5 timed repetitions per clip. For the subset of nine clips that passed through the entire verification pipeline (45 runs in total), the client-side latency measured from the end of the utterance was 224.3 ± 21.7 ms. The average server-side latency for the same clips was 223.7 ± 21.6 ms, with a range of 184.4–262.0 ms. The first, uncached session is reported separately and took 1875.1 ms. The clip set is small and repetitions are deterministic repeats, not independent samples. A local in-process test client was used with no network transmission, so these figures exclude network delay. The PyTorch models ran on Apple Silicon MPS (torch.device("mps")).

On average, SecureKYC AI returns its verdict about 0.22 s after the utterance ends once the models are warm (the first session took 1.9 s), within our own sub-second target.

D. Distinctions from Closest Prior Work

SecureKYC AI differs structurally from related work. Agarwal et al. [18] analyzed bilabial closures on pre-recorded speech using open-domain transcription, whereas SecureKYC AI issues the prompt itself, aligns audio to it with MMS_FA forced alignment, and adds a normalized 3D aperture metric and constrained late fusion. GOTCHA [9] relies on visual challenges (hand-on-face occlusion, standing up, head yaw; 80.1% automated and 88.6% human AUC) and does not use voice; SecureKYC AI uses the applicant's speech instead of gestures. LipFD [19] and PIA [21] detect audio-visual inconsistency with a vision transformer backbone and a WhisperX/ArcFace-based 3D CNN, respectively; SecureKYC AI instead tests directly for the physical bilabial seal ($D_lip \leq \tau$) against a known transcript, with a post-utterance latency of 224.3 ± 21.7 ms on Apple Silicon MPS (local client, excluding network delay). BioLip [20] models biomechanical lip jitter without audio, whereas SecureKYC AI matches 3D lip-aperture dynamics to acoustic forced alignment.

E. Limitations and Negative Results

Our empirical findings are subject to several explicit boundary conditions and negative results documented in the accompanying repository:

1) Cross-dataset and in-distribution bounds: Verification performance is established strictly in-distribution on the VidTIMIT and DeepFakeTIMIT benchmarks (deployed model: 68.93% accuracy, 0.7197 test ROC-AUC, 36.02% EER, with representative 5-fold cross-validation generalization of 0.6877 ± 0.0449 and 36.00 ± 2.98% EER). At the default threshold the deployed model fails to catch 22 out of 46 attack videos (APCER 47.83%), which falls far below the level necessary for a real-world KYC application. The closure threshold τ was tuned on 15 VidTIMIT subjects, including two from the test split (fadg0, fcmr0).
2) Negative-result dataset evaluations: Three third-party datasets did not produce a cross-modal desynchronization signal. FakeAVCeleb v1.2 (N=50 pilot, 11 fully aligned samples, with 56% forced-alignment attrition on free-form speech) resulted in $S_{desync} = 0.0000$ and $d = -0.426$, the opposite direction, where fake videos contained fewer closure failures than real clips, thereby triggering the rejection decision gate ($d < 0.50$). The 8-subject offline InsightFace inswapper dataset showed negligible separation ($d = 0.27$), since 2D face swapping keeps the kinematic features of the speaking actor in each frame. Celeb-DF v2 has no audio track (N=0 bilabial events), and a 1D visual aperture baseline yielded near-chance test AUC (0.5111).
3) Live injection validation: An active OBS Virtual Camera loopback test with a live DeepFaceLive model (.dfm) was not completed and remains future work.
4) Acoustic Prompt Verification Rejection Gate (CER Gate): The gating threshold 0.60 was selected as the empirical midpoint of the separation gap observed in a preliminary 15-clip sweep and is not a validated or secure boundary. On N=335 genuine VidTIMIT clips with canonical transcripts, decoded with Wav2Vec2 CTC, genuine CER has median 0.372, mean 0.381, 95th percentile 0.873 and maximum 1.000; 51 clips (15.2%) exceed 0.60. When evaluated against all five active challenge prompts (335 clips x 5 prompts = 1,675 mismatched pairs), mismatch CER exhibits min 0.6471, 5th percentile 0.7632, and median 0.8409, yielding a false-accept rate of 0.00% (0/1,675 below 0.60). Thus, at threshold 0.60, the genuine clip false rejection rate is 15.2% (51/335) beside a mismatched prompt false acceptance rate of 0.00% (0/1,675). The mismatches are real speech scored against prompts that were not spoken; the gate does not address an attacker who speaks the correct phrase.

VI. CONCLUSION & FUTURE WORK

In this paper, we introduced SecureKYC AI, an active challenge-response biometric defense architecture for remote Video KYC that targets the weakness of passive vision checks against virtual-camera injection and live human puppeteers. By exploiting the physical requirement of lip closure in bilabial speech ($P_{oral} >> P_{ambient}$ in /m/, /b/, /p/) and coupling 3D lip aperture dynamics with MMS_FA trellis forced alignment, SecureKYC AI separates genuine from reenacted speech on DeepFakeTIMIT without open-vocabulary ASR, at a moderate accuracy level (Section V-E).

Evaluated on N=691 audio-visual samples across 43 disjoint identities with zero synthetic data, our deployed constrained late-fusion meta-learner delivers 68.93% accuracy, 70.59% precision, 52.17% recall, 0.7197 test ROC-AUC (with a 5-fold cross-validation generalization estimate of 0.6877 ± 0.0449), 0.6000 F1-score, and 36.02% EER. At the default 0.5 operating point, late fusion produced 10 false alarms on the 57 genuine test clips versus 16 for the standalone vision branch (paired bootstrap 95% CI for the difference [0, 12]), with identical recall (52.17%) and higher precision (70.59% vs 60.00%); at the equal-error point the two models are identical. Verification completes in 224.3 ± 21.7 ms after the utterance ends (9 clips, 45 runs, local in-process client). Future work will investigate per-subject resting aperture calibration during enrollment to further enhance cross-speaker generalizability across unconstrained populations.

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